

Tribhuvan University
Institute of Science and Technology
2065

Bachelor Level/ First Year/ First Semester/ Science
Computer Science and Information Technology (CSc. 113)
(Physics)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Section A:

Attempt any four questions:

1. What is meant by Galilean invariance? Show that distance and acceleration are invariant to Galilean transformation, velocity is not invariant. (2+1.5+1.5+2)
2. It is given that the potential energy of a system is rotationally invariant. What do you mean by rotational invariance? Show that angular momentum is conserved for such a system. (3.5+3)
3. (a) Discuss the analogy between liquid-flow and current-flow and hence, derive an expression for liquid-flow through capillaries in series. (4)
(b) State Gauss's law and use it to show that excess charge of a charged conductor resides on its outer surface. (3)
4. (a) Discuss the analogy between liquid-flow and current-flow and hence, derive an expression for liquid-flow through capillaries in series. (4)
(b) State Gauss's law and use it to show that excess charge of a charged conductor resides on its outer surface. (3)
5. Derive the expression for energy density in the magnetic field. (7)
6. Explain the empirical basis for writing the Maxwell's equations and write them. (7)

Section B:

Attempt any eight questions:

7. A proton is accelerated through a potential difference 50V and then it is allowed cross a field free region 7.5m long. Find the time required to cross this distance. (4)
8. Find the height of geostationary satellite (as viewed by an observer on the earth's surface); given $g=9.8 \text{ ms}^{-2}$ on the earth's surface, $R= 6.38 \times 10^5 \text{m}$. (4)
9. The potential energy for the Vander Waals force between two atoms is given by $U(X) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where x is the distance between the atoms and a and b are positive constants. Calculate the force between the two atoms and plot it against x. (4)
10. A parallel LCR circuit has $L= 8\text{mH}$, $C= 10 \mu\text{F}$ and $R= 0.5\Omega$. Calculate the natural frequency and quality factor. (4)
11. A water drop of radius 0.01 mm is falling through air neglecting the density of air as compared to the water, calculate the terminal velocity of the drop (η for air = 1.8×10^{-4} CGS units) (4)
12. Two point charges of $\frac{q}{2}$ are located at the origin and at (a, 0, 0) respectively. Find the point where electric field vanishes. (4)
13. Two parallel conducting plates are separated by the distance d and potential difference $\Delta\psi$. A dielectric slab of dielectric constant k is and of uniform thickness is tightly fitted between the plates. Find the electric field in the dielectric. (4)
14. What is the capacitance of a capacitor that can store 800 J at 800 V? Suppose the capacitor has parallel plates separated by 10^{-5} m and filled with a dielectric of dielectric constant 2.2. What is the area of the plates? (4)

15. Consider a simple RL circuit in which a sudden voltage V is applied. Discuss its transient behavior and find the current as a function of time. (4)
16. Show that the time average power dissipation in a circuit which carries an AC current $i(t) = i_0 e^{i\omega t}$ and $V(t) = z i(t)$ is $\bar{P} = \frac{1}{2} |i_0| |V_0| \cos \theta$. Here z is the impedance of the circuit:
 $Z = |Z| e^{i\theta}$ (4)